COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:		**SECTION:**	CPE11S4
DATE PERFORMED:	Aug 5 20026	**DATE SUBMITTED:**	Aug 5 2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

1)

```
#include <iostream>
using namespace std;

int main() {
    int choice;
    int Num;
    float balance, charges, credits, limit, newBalance;

    do {
    cout << "\n1 = Enter a Account number" << endl;
    cout << "0 = Quit" << endl;
    cout << "Your choice: ";
    cin >> choice;

        switch (choice) {
        case 1:
            cout << "Enter account number: ";
            cin >>Num;
            cout << "Enter balance: ";
```

```
    cin >> balance;
    cout << "Enter charges: ";
    cin >> charges;
    cout << "Enter credits: ";
    cin >> credits;
    cout << "Enter limit: ";
    cin >> limit;

    newBalance = balance + charges - credits;
    cout << "New Balance: " << newBalance << endl;

    if (newBalance > limit) {
        cout << "expired" << endl;
    } else {
        cout << "good" << endl;
    }
    break;

case 0:
```

Output

```
1 = Enter a Account number
0 = Quit
Your choice: 1
Enter account number: 2877
Enter balance: 6000
Enter charges: 1200
Enter credits: 900
Enter limit: 5000
New Balance: 6300
expired
```

```
1 = Enter a Account number
0 = Quit
Your choice: 1
Enter account number: 399
Enter balance: 300
Enter charges: 10
Enter credits: 200
Enter limit: 500
New Balance: 110
good
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1) Because of the price of gasoline, drivers are concerned with the mileage obtained by their automobiles. One driver has kept track of several tankfuls of gasoline by recording miles driven and gallons used for each tankful. Develop a program that will input the miles driven and gallons used for each tankful. The program should calculate and display the miles per gallon obtained for each tankful. After processing all input information, the program should calculate and print the combined miles per gallon obtained for all Tankfuls

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int miles, tankful, all;
    float gallon;
    cout << "Enter the mileage: ";
    cin >>miles;
    cout << "How much gallon: : ";
    cin >>gallon;
    cout << "How many tankful: ";
    cin >>tankful;

    all = miles + gallon * tankful;
    cout << "total result: " << all << endl;

    return 0;
}
```

Output

```
Enter the mileage: 1500
How much gallon: : 3.785
How many tankful: 2
total result: 1507


=== Code Execution Successful ===
```

2) Create a program that will calculate the cost of sending a small parcel. The post office charges P5.00 for the first 300g, and P2.00 for every 100g thereafter (rounded up), up to a maximum weight of 1000g.

main.cpp

```cpp
#include <iostream>
using namespace std;

int main() {
    int weight;
    double cost;

    cout << "Enter the weight: ";
    cin >> weight;

    if (weight <= 0 || weight > 1000) {
        cout << "Invalid!" << endl;
    } else {
        if (weight <= 300) {
            cost = 5.00;
        } else if (weight <= 900) {
            cost = 5.00 * 2.00;
        } else {
        }

        cout << "Total cost: " << cost << endl;
    }

    return 0;
}
```

Output

```
Enter the weight: 299
Total cost: 5


=== Code Execution Successful ===
```

Output

```
Enter the weight: 600
Total cost: 10


=== Code Execution Successful ===
```

Output

```
Enter the weight: 0
Invalid


=== Code Execution Successful ===
```

3)Write a program that displays a menu for simple conversions such as the following:
(1) cm - inches
(2) inches - cm
(3) feet - meter
(4) meter - feet
Once selected, the user will be asked to enter a float to be converted. After the conversion, the user would be asked to convert another until the user doesn't want any more. Display your samples for all conversions

main.cpp

Share

```cpp
int main() {
    int choice;
    float value;
    char again;

    do {
        cout << "1 = cm to inches\n";
        cout << "2 = inches to cm\n";
        cout << "3 = feet to meter\n";
        cout << "4 = meter to feet\n";
        cout << "Choice: ";
        cin >> choice;
        cout << "Enter value: ";
        cin >> value;

        switch(choice) {
            case 1:
            cout << value << " cm = " << value / 2.54 << " inches" << endl;
            case 2:
            cout << value << " inches = " << value * 2.54 << " cm"<< endl;
            case 3:
            cout << value << " feet = " << value * 0.3048 << " meters"<< endl;
            case 4:
            cout << value << " meters = " << value / 0.3048 << " feet"<< endl;
                break;
            default:
                cout << "Invalid Choice"<< endl;
        }
        cout << "again?: ";
        cin >> again;
    } while (again == 'G' || again == 'G');

```

Output

```
1 = cm to inches
2 = inches to cm
3 = feet to meter
4 = meter to feet
Choice: 1
Enter value: 25
25 cm = 9.84252 inches
25 inches = 63.5 cm
25 feet = 7.62 meters
25 meters = 82.021 feet
again?: G
1 = cm to inches
2 = inches to cm
3 = feet to meter
4 = meter to feet
Choice: 2
Enter value: 90
90 inches = 228.6 cm
90 feet = 27.432 meters
90 meters = 295.276 feet
again?: G
1 = cm to inches
2 = inches to cm
3 = feet to meter
4 = meter to feet
Choice: 5
Enter value: 4
Invalid Choice
again?:
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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